Aastik Mishra

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Education

Birla Institute of Technology, Mesra *July 2019 – July 2023*
B.Tech in Computer Science and Engineering CGPA: 8.16

Experience

Oracle | Software Engineer - MTS Jun 2025 - Present

- Contributed to **Oracle ACFS (cluster filesystem)** validation infrastructure, working on backend test frameworks for storage and volume management systems.
- Designed and implemented automated workflows to validate **volume, snapshot, clone, and quarantine operations**, ensuring correctness under concurrent and failure scenarios.
- Analyzed large-scale test failures by parsing logs, kernel outputs, and .dif artifacts, performing detailed root cause analysis across filesystem and system layers in production-like environments.
- Improved observability by introducing structured logging (Tlog) and enhancing test execution visibility, significantly reducing debugging time for complex infra-level failures across multiple test scenarios.
- Enhanced reliability of test execution pipelines by refining command orchestration and error-handling logic in **Perl-based** backend frameworks, improving overall execution stability and consistency.

Unique Identification Authority of India | Software Developer - Full Stack July 2023 - May 2025

- Designed and implemented a high-throughput **Go-based microservice** for Aadhaar reactivation using **gRPC** and HTTP, handling large-scale concurrent requests with optimized latency and reliability.
- Built a **distributed enrolment tracking system** using **Go** and **MySQL**, enabling real-time status updates and improving system observability through **OpenTelemetry (OTEL)** instrumentation.
- Improved system reliability and debugging by integrating **centralized logging, metrics, and tracing**, reducing issue resolution time during production incidents.
- Developed and deployed services using **CI/CD pipelines (ArgoCD)**, ensuring automated rollouts and environment consistency across deployments.
- Implemented secure authentication and communication using **JWT**, **gRPC**, and encryption techniques, ensuring safe data exchange in high-volume identity systems.
- Reduced manual processing overhead by **40%** by integrating child enrolment workflows (BRN, DigiLocker, Passport) into existing backend systems.
- Built a React-based authentication interface integrating biometric verification flows, improving system usability.

Wabtec Corporation | Software Engineer Intern Jan 2023 - Jun 2023

- Developed an automated routing system using **C++** and **JavaScript**, optimizing execution plans and reducing manual intervention by **30%**.
- Implemented real-time signal clearance logic based on movement constraints, improving system efficiency and decision accuracy.

Bytelearn | Software Engineer Intern Dec 2022 - Jan 2023

- Improved frontend performance and responsiveness using **React.js** and **Node.js**, enhancing user engagement.
- Identified and resolved cross-device compatibility issues, improving overall platform stability.

Projects

High-Throughput Order Management System Go Project - Jan 26

- Built a high-performance trading engine simulating brokerage systems with order placement, validation, & execution workflows.
- Implemented Redis-backed order book & state management for fast reads/writes & concurrent consistency.
- Designed an event-driven architecture using Kafka for asynchronous processing and WebSockets for real-time order updates.

Okta Authentication Flutter Project - Aug 24

- Implemented secure authentication using **OAuth 2.0** and **Okta**, enabling cross-platform login for Android/iOS applications.
- Designed secure token handling and session management for mobile environments.

XML Signer React App - Dec 23

- Built a tool to digitally sign XML/JSON payloads, ensuring **data integrity** and **tamper-proof transmission**.
- Implemented cryptographic validation workflows for secure data exchange systems.

Technical Skills

Languages: Java, Go, Perl, C/C++, SQL, HTML, CSS, JavaScript • **Tools and Frameworks:** React.js, Spring Boot, Node.js, gRPC, REST APIs, Gin, MongoDB, DynamoDB, Filesystems (ACFS), MySQL, AWS, OAuth, JWT, ArgoCD, Bootstrap, Tailwind, JQuery, Sass, Express, Git, Postman, GitHub, VS Code **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Object Oriented Programming, Networking, Database Management Systems, System Design